# AI COMIC STRIP GENERATOR

## PROJECT DOCUMENT

**From one idea to a finished comic: story by LLM, art by Stable Diffusion, export as PDF!**

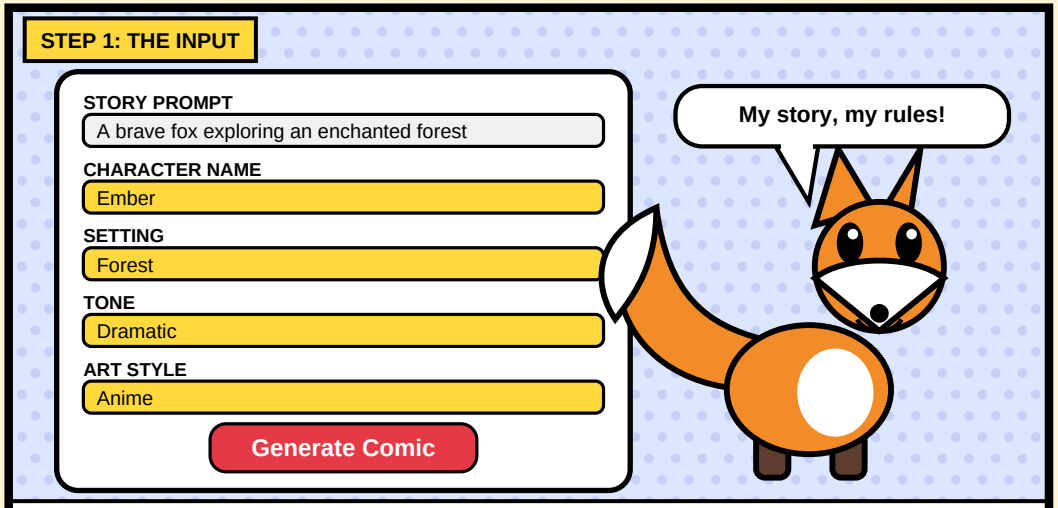



**INSIDE:**

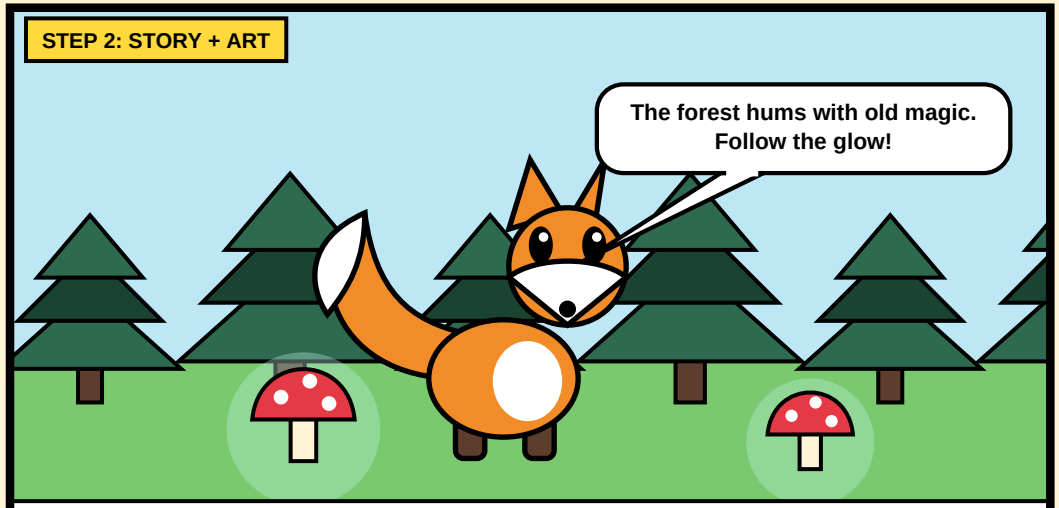
* Scenario 1 - Personalized comic from a single prompt
* Scenario 2 - Re-styling tone and art on demand
* Scenario 3 - Layout binding and PDF export
* System pipeline and feature summary

# SCENARIO 1: A STORY FROM ONE PROMPT

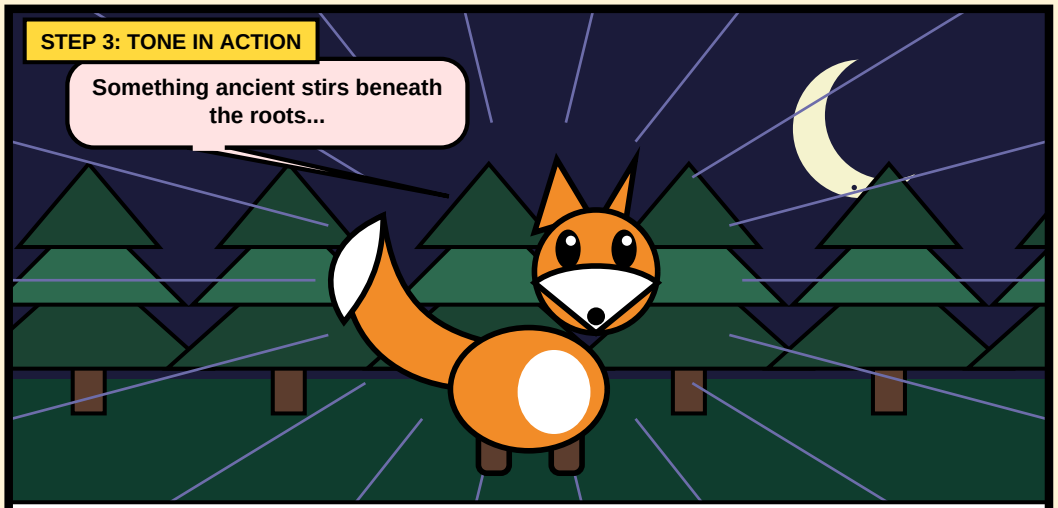
*Personalized story generation with LLM + Stable Diffusion*



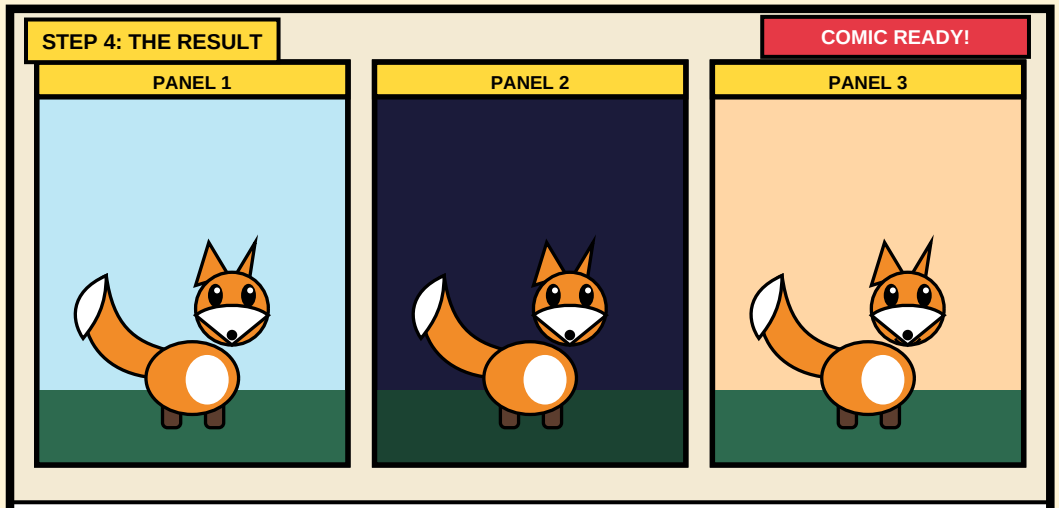

The user types the idea and picks preferences: character name, setting, tone and art style. Everything is captured on one input form.




The LLM turns the prompt into an outline and panel-by-panel story. Stable Diffusion then paints each scene in the chosen style.




Tone and style flow into every prompt, so a dramatic, anime-style forest reads as one consistent world from panel to panel.

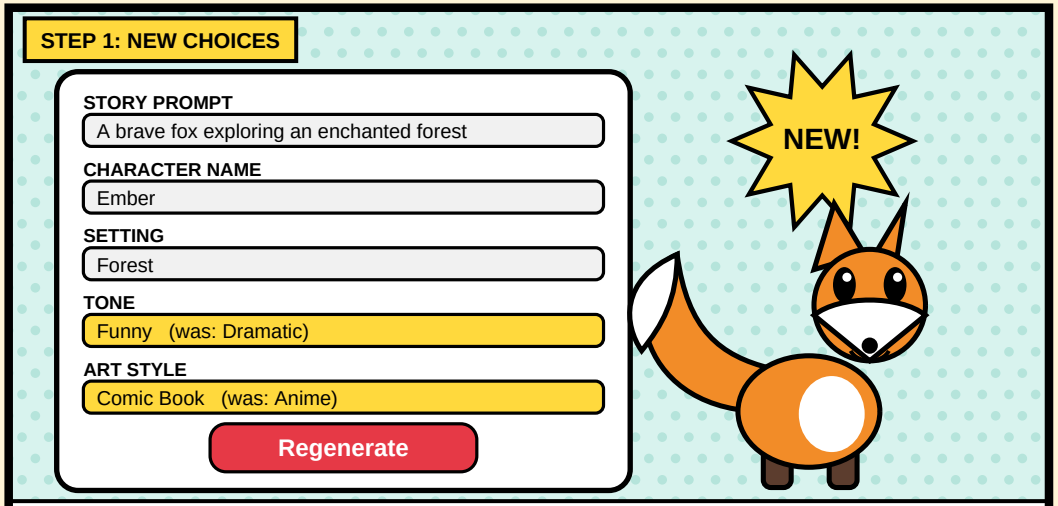



Multiple panels, each with an image and matching text, form a cohesive comic strip that fits the creator's vision and boosts satisfaction.

# SCENARIO 2: CHANGE THE MOOD, REGENERATE

*Customize tone and art style, then rerun the whole pipeline*

**STEP 1: NEW CHOICES**




Instead of the default or previous picks, the user selects Funny for tone and Comic Book for art style, then submits the form again.

**STEP 2: FULL REGENERATION**

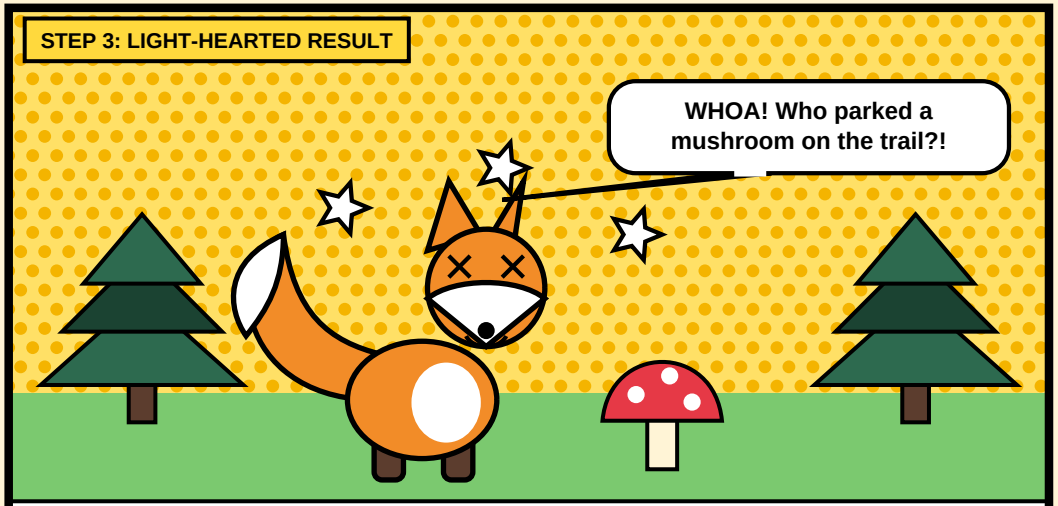
**FULL PIPELINE RE-RUNS WITH NEW SETTINGS**




*tone = funny | style = comic book*

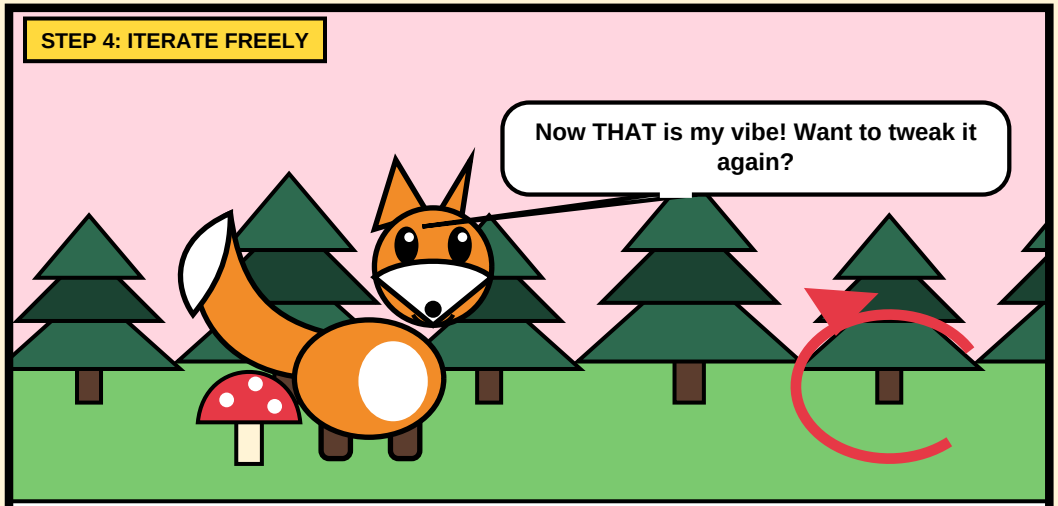
The system reruns the entire pipeline (outline, story and images) using prompts and model instructions tailored to the new settings.

**STEP 3: LIGHT-HEARTED RESULT**




A humorous narrative with a classic comic-book look: bold outlines, halftone dots and playful, cartoonish expressions.

**STEP 4: ITERATE FREELY**

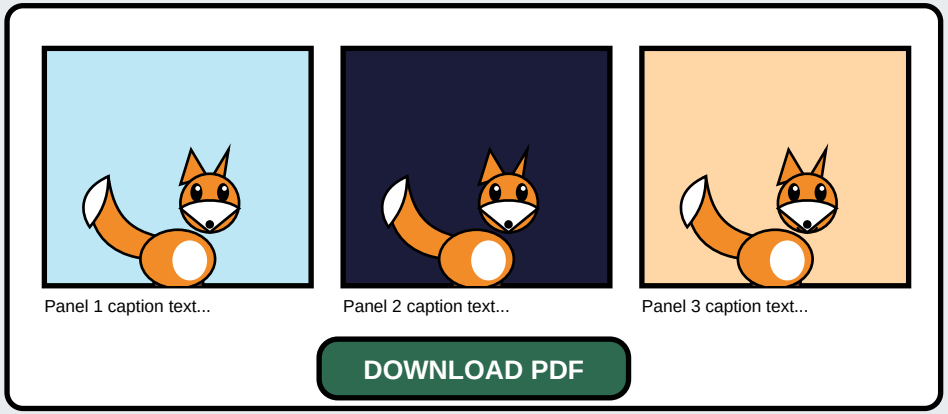



Users can keep adjusting mood and visual style, again and again, until the comic matches their creative vision.

# SCENARIO 3: PREVIEW, EXPORT, DOWNLOAD

*Layout binding and PDF generation*

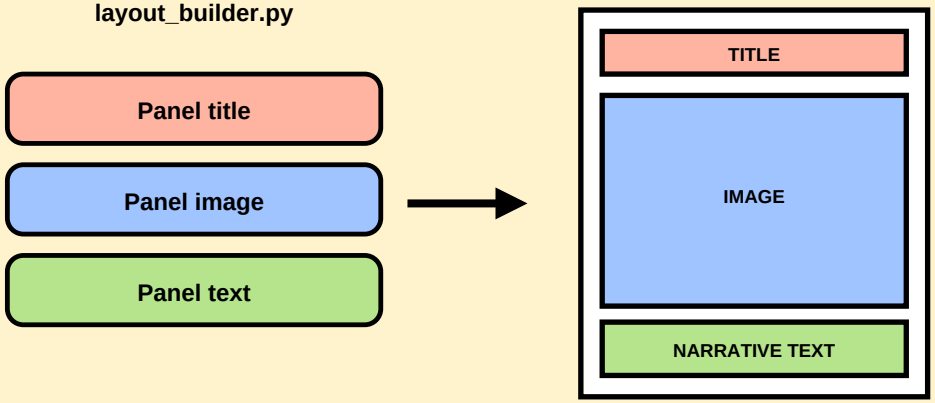
**STEP 1: REVIEW + DOWNLOAD**




After checking the on-screen preview, the user is happy with the comic and clicks the download button.

**STEP 2: LAYOUT BINDING**

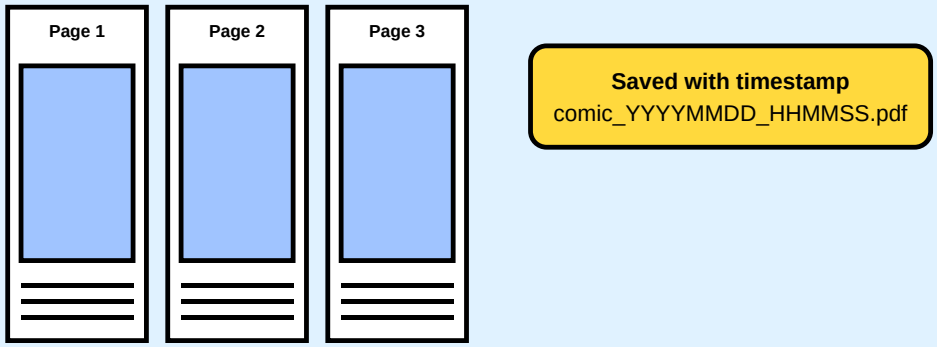
**layout_builder.py**




layout_builder.py assembles each panel's title, image and text into a structured layout, ready for export.

**STEP 3: PDF GENERATION**

**exporters.py + FPDF**




exporters.py compiles the comic with FPDF, placing images and narrative text on consecutive pages, then saves a timestamped PDF.

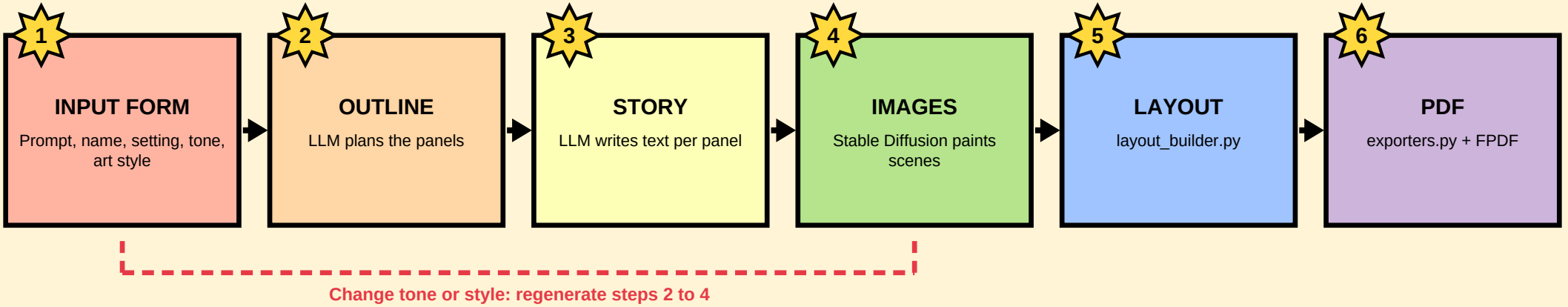
**STEP 4: EXPORT SUCCESS**




The user is redirected to the export success page, confirming the download. The comic can be saved, printed or shared professionally.

# THE BIG PICTURE

*System pipeline and key benefits*




| PERSONALIZED | ITERATIVE | COHESIVE | SHAREABLE |
|---|---|---|---|
| Name, setting, tone and art style shape every panel, so the comic matches the creator's vision. | Switch from dramatic anime to funny comic book in one submit and compare results. | Story text and images are generated from shared settings, keeping panels consistent. | A timestamped, professional PDF that is easy to save, print or share. |

